

MARGDARSHAK AI

"From Classroom to Dream Career — Guided by AI"

Founder & Lead Developer: Mehak

Edition & Track: SIH 2024 Edition

Live Platform: margdarshak-ai-khaki.vercel.app

GitHub: github.com/Mehak237/margdarshak-ai

1. Executive Summary

In India's higher education ecosystem, over **1.5 million engineers and degree graduates** enter the job market annually. However, national surveys (including the Wheebox India Skills Report) reveal that **less than 45% of graduating students from Tier-2 and Tier-3 colleges** possess industry-ready technical, behavioral, and communication proficiencies. The root cause is an acute **information and mentorship divide**: students lack insight into company-specific ATS expectations, miss out on thousands of crores in unutilized scholarships, and cannot afford expensive private placement coaching.

Core Project Innovation: Margdarshak AI is an end-to-end, zero-cost, browser-based career co-pilot architected and developed by **Mehak**. It bridges this gap through instant **Resume vs. Dream Job Gap Analysis** across 60+ Indian employers, **25+ verified scholarships** with an AI Statement of Purpose (SOP) writer, **real-time interactive voice AI mock interviews** with instant scoring, and complete **multilingual support in 9 Indian languages**.

2. Problem Statement & Ground Realities in Tier-2/3 Colleges

- 1. The Vague Job Description & ATS Trap:** Students submit generic resumes lacking company-specific core keywords, leading to automated ATS disqualification before human review.
- 2. Massive Scholarship Information Asymmetry:** Over Rs. 2,000 Crores in central, state, and corporate scholarship funds go unused because students are unaware of eligibility or cannot draft strong Statements of Purpose.
- 3. Exorbitant Private Coaching Fees:** Commercial interview coaching services charge Rs. 1,500 - Rs. 3,000 per session, completely out of reach for students from humble economic backgrounds.
- 4. The Linguistic Divide:** Standard technical career coaching is almost exclusively in English, triggering severe anxiety among vernacular-speaking students during campus recruitment drives.

3. Technical Architecture & Hybrid Edge Design

Margdarshak AI adopts a **Hybrid Edge-First Architecture**. It runs 100% client-side by default—delivering **0ms cold starts**, total student data privacy, and zero operational server costs—while offering progressive cloud enhancement via Google Gemini 1.5 Flash and multi-user database adapters.

Component Layer	Technologies Used	Technical Responsibility
Frontend UI/UX	Vanilla HTML5, Tailwind CSS, Lucide Icons	Semantic single-page application with responsive dark/light glassmorphism interface.
Resume Engine	PDF.js Client-Side Worker	Extracts text locally in browser memory without sending private CVs to external servers.
Voice Interviewer	Web Speech Synthesis & SpeechRecognition API	Vocalizes questions and captures spoken answers through microphone in real time.

Offline / PWA	Service Worker (sw.js) & Web App Manifest	Caches assets locally for low-bandwidth operation on unstable college networks.
AI Engine	Google Gemini 1.5 Flash + Local Heuristics	Dual-mode: deterministic offline rule engine plus dynamic Gemini generative evaluation.
Backend & Cloud	Node.js REST API (server.js) / Supabase / MongoDB	Cloud database sync for multi-user student profiles and scan history persistence.
Hosting & Edge	Vercel Global Edge Network	Global CDN distribution with automated CI/CD directly from GitHub repository.

4. Core Features Deep Dive — Built by Mehak

4.1 Resume vs. Dream Job Gap Analyzer & 15-Day Roadmap

- **60+ Indian Tech Employers:** Encompasses Product Giants (Google, Amazon, Microsoft, Adobe), IT Titans (TCS Ninja & Digital, Infosys SE & DSE, Wipro, Cognizant, LTIMindtree), E-commerce/Unicorns (Flipkart, Swiggy, Zomato, CRED), and Fintech (Goldman Sachs, Morgan Stanley).
- **Algorithmic ATS Matching:** Compares applicant skills against company benchmarks, separating *Critical Must-Have Skills* from *Bonus Competitive Skills*.
- **Dynamic SVG Progress Meter:** Real-time SVG circular meter with reactive color grading (Green >=80%, Amber 60-79%, Rose <60%).
- **Automated 15-Day Action Roadmap:** Phase 1 (Days 1-5: Foundation repair), Phase 2 (Days 6-10: Hands-on project building), Phase 3 (Days 11-15: Mock interviews and system design).
- **Pitch Evaluator:** NLP evaluation of candidate's 60-second 'Tell Me About Yourself' elevator pitch.

4.2 Smart Scholarship Engine & AI SOP Co-Pilot

- **25+ Verified Indian Schemes:** Curated database including AICTE Pragati (Rs. 50,000/yr for girls), Reliance Foundation (Rs. 2,00,000), Tata Trust, NSP Central Sector, Post-Matric SC/ST, and ONGC Scholars.
- **6-Dimensional Real-Time Filter:** Filters simultaneously by Family Income (Rs. 1.5L to Rs. 8L EWS ceiling), Academic Percentage (40% to 98% slider), Social Category, Gender, Domicile State, and Degree Level.
- **Committee-Ready AI SOP Co-Pilot:** Generates professional 4-paragraph Statements of Purpose tailored to selection committees, emphasizing academic passion, financial resilience, and future commitment to society.

4.3 Real-Time Voice AI Mock Interviewer with Live Scorecard

- **200+ Company-Specific Question Bank:** Mapped to live interview formats of TCS, Infosys, Amazon, etc.
- **Speech Synthesis (Interviewer Voice):** Vocalizes questions aloud using HTML5 Web Speech Synthesis API.
- **Speech Recognition (Voice Mic Input):** Captures candidate speech via microphone, testing live verbal clarity.
- **10-Point Scorecard Rubric:** Evaluates answers on Technical Accuracy (4 pts), Structure (2 pts), STAR Method (2 pts), and Confidence (2 pts), with actionable constructive feedback.

4.4 Multilingual Support (9 Indian Languages) & PWA Offline Engine

- **9 Indian Languages:** Full zero-reload DOM localization in English, Hindi, Hinglish, Tamil, Telugu, Kannada, Bengali, Marathi, and Gujarati.
- **PWA Offline Mode:** Service worker (sw.js) and manifest (manifest.json) allow full offline installation and usage.
- **1-Click WhatsApp Batch Sharing:** Pre-formatted actionable invitation card enabling viral distribution across college batch groups.

5. Mathematical ATS Match Formulation

The match score between candidate resume text and target company profile is calculated using a dual-weighted polynomial:

ATS Compatibility Score Formula:
Score_ATS = [(Matched_Core_Skills / Total_Core_Skills) x 70%] + [(Matched_Bonus_Skills / Total_Bonus_Skills) x 30%]

- **Score >= 80% (Green Tier):** Placement Ready — Strongly aligned for immediate application.
- **Score 60% - 79% (Amber Tier):** Competitive with Gaps — Requires 5-10 days targeted upskilling.
- **Score < 60% (Rose Tier):** Fundamental Foundations Required — Follow Phase 1 Roadmap.

6. Codebase Structure & File Responsibilities

File Path	Module Type	Key Technical Functionality
index.html	SPA Layout	Single Page Application UI, responsive navigation, modals, and metric counters.
css/style.css	Design Tokens	Glassmorphism styles, dark/light theme tokens, keyframe animations, glow effects.
js/app.js	Master Controller	Coordinates tab switching, theme toggling, share modals, and Lucide icons.
js/gapAnalyzer.js	Gap Engine	Client-side PDF.js parsing, 60+ company skill matcher, dynamic SVG score ring.
js/scholarshipEngine.js	Scholarship Engine	6-dimensional filter engine and committee-ready 4-paragraph AI SOP co-pilot.
js/mockInterviewer.js	Voice Interviewer	Web Speech API voice synthesis, microphone STT, and 10-point scorecard rubric.
js/auth.js	Authentication	Universal document-delegated authentication, session handling, demo student seeds.
data/companies.js	Dataset	60+ Indian employers mapped across 5 tiers with core and bonus skill taxonomies.
data/scholarships.js	Dataset	25+ verified central, state, and corporate scholarships with eligibility rules.
data/translations.js	Localization	Multilingual key-value dictionary for 9 Indian languages.
server.js	Full-Stack API	Native Node.js REST API with zero external dependencies and cloud DB sync.
sw.js & manifest.json	PWA Engine	Service worker cache-first offline strategy and standalone web app installation.
vercel.json	Edge Routing	Static edge routing configuration for zero-downtime production deployment.

7. Social Impact, NEP 2020 Alignment & Future Roadmap

- **NEP 2020 Alignment:** Directly furthers national educational directives by replacing theoretical career ambiguity with verified industry skill benchmarks and vernacular accessibility.
- **Economic Empowerment:** Upgrading a candidate from a service baseline (Rs. 3.5 LPA) to a product/digital tier (Rs. 7.5 LPA) increases lifetime earning potential by over Rs. 25 Lakhs per student.
- **Future Roadmap:** Phase 2 (Q4 2026) will introduce a College Placement Officer (TPO) analytics portal and an in-browser DSA code runner. Phase 3 (2027) will add peer-to-peer mock interview rooms.

PROJECT CERTIFICATION & ACADEMIC EVALUATION

This technical report certifies that **Margdarshak AI** has been fully conceptualized, architected, engineered, and deployed live to production by **Mehak** as an original technological innovation for the **Smart India Hackathon (SIH 2024 Edition)**.

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Faculty / Mentor Signature
Academic Project Evaluator / SIH Committee
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